VEHICLE INSURANCE DATASE

Presented by

Rahul Sharma

rahsharmaz465@gmail.com

(EDA) on Vehicle Insurance Dataset

Exploratory Data Analysis (EDA) on a Vehicle Insurance dataset helps in understanding patterns, trends, and relationships among different variables affecting insurance policies and claims. The dataset usually includes features like age, gender, vehicle type, premium amount, policy type, and claim status.

The process begins with data cleaning, where missing values, duplicates, and incorrect data types are handled. Univariate analysis is used to study individual variables through histograms and count plots. For example, it helps understand the distribution of age or the frequency of different vehicle types.

Bivariate analysis is then performed to explore relationships between variables, such as how age or driving history impacts premium costs or claim likelihood. Visual tools like bar charts, box plots, and heatmaps make insights clearer.

Overall, EDA helps identify risk factors, improve pricing strategies, and support better decision-making in the vehicle insurance domain.

This dataset contains 10 attributes with simple meaning and which are described as follows:

Age
Represents the age of the customer. It helps in analyzing how age impacts insurance purchase and claim patterns.
Gender
Indicates whether the customer is male or female. Useful for comparing claim frequency and policy preferences.
Annual Premium
The amount paid by the customer yearly for the insurance policy. It reflects the cost of coverage chosen.
Policy Type
Describes the type of insurance policy selected (e.g., basic, comprehensive). It shows coverage level.
Vehicle Age
Indicates how old the vehicle is (e.g., less than 1 year, 1–2 years, more than 2 years). Older vehicles may have higher risk.
Vehicle Type
Specifies the category of vehicle (e.g., car, bike, commercial). Different vehicles carry different risk levels.
Driving License
Shows whether the customer has a valid driving license (Yes/No). Important for legal and risk assessment.
Previously Insured
Indicates if the vehicle was insured before. Helps understand customer history and continuity.
Claim Status
Represents whether the customer has made a claim (Yes/No). This is a key target variable for analysis.
Policy Duration
Shows the duration of the insurance policy (e.g., 1 year, 2 years). Helps analyze when claims are most frequent.

IMPORT LIBRARIES

- Required libraries were imported for data analysis
- Pandas and Numpy for data handling
- Matplotlib and Seaborn for visualization

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Data Loading

- The dataset was load into a Pandas Dataframe.
- CSV file was imported successfully for analysis
- Data is now ready for further exploration and processing

```
from google.colab import drive
drive.mount('/content/drive')
```

```
path='/content/drive/MyDrive/Vehicle_Insurance.csv'
df=pd.read_csv(path)
df
```

	id	Gender	Age	Driving_License	Region_Code
0	1	Male	44	1	28.0
1	2	Male	76	1	3.0
2	3	Male	47	1	28.0
3	4	Male	21	1	11.0
4	5	Female	29	1	41.0
...	...	...	...	...	...
381104	381105	Male	74	1	26.0
381105	381106	Male	30	1	37.0
381106	381107	Male	21	1	30.0
381107	381108	Female	68	1	14.0
381108	381109	Male	46	1	29.0

381109 rows × 12 columns

Data Description

- Generated statistical summary of numraical features
- Includes mean,menimum, maximum, and standard deviation
- Helps in understanding data distribution and range

```
df.describe()
```

	id	Age	Driving_License
count	381109.000000	381109.000000	381109.000000
mean	190555.000000	38.822584	0.997869
std	110016.836208	15.511611	0.046110
min	1.000000	20.000000	0.000000
25%	95278.000000	25.000000	1.000000
50%	190555.000000	36.000000	1.000000
75%	285832.000000	49.000000	1.000000
max	381109.000000	85.000000	1.000000

Data information

- Checked data types of all columns
- Identified non-null values
- Helped in understanding data structure

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 381109 entries, 0 to 381108
Data columns (total 12 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   id                    381109 non-null  int64
 1   Gender                381109 non-null  object
 2   Age                   381109 non-null  int64
 3   Driving_License       381109 non-null  int64
 4   Region_Code           381109 non-null  float64
 5   Previously_Insured    381109 non-null  int64
 6   Vehicle_Age           381109 non-null  object
 7   Vehicle_Damage        381109 non-null  object
 8   Annual_Premium        381109 non-null  float64
 9   Policy_Sales_Channel  381109 non-null  float64
 10  Vintage               381109 non-null  int64
 11  Response              381109 non-null  int64
dtypes: float64(3), int64(6), object(3)
memory usage: 34.9+ MB
```

Missing Value Detection

- Use isnull().sum() function to examine dataset structure
- Checked non-null values for each column
- Confirmed that missing values were handled properly

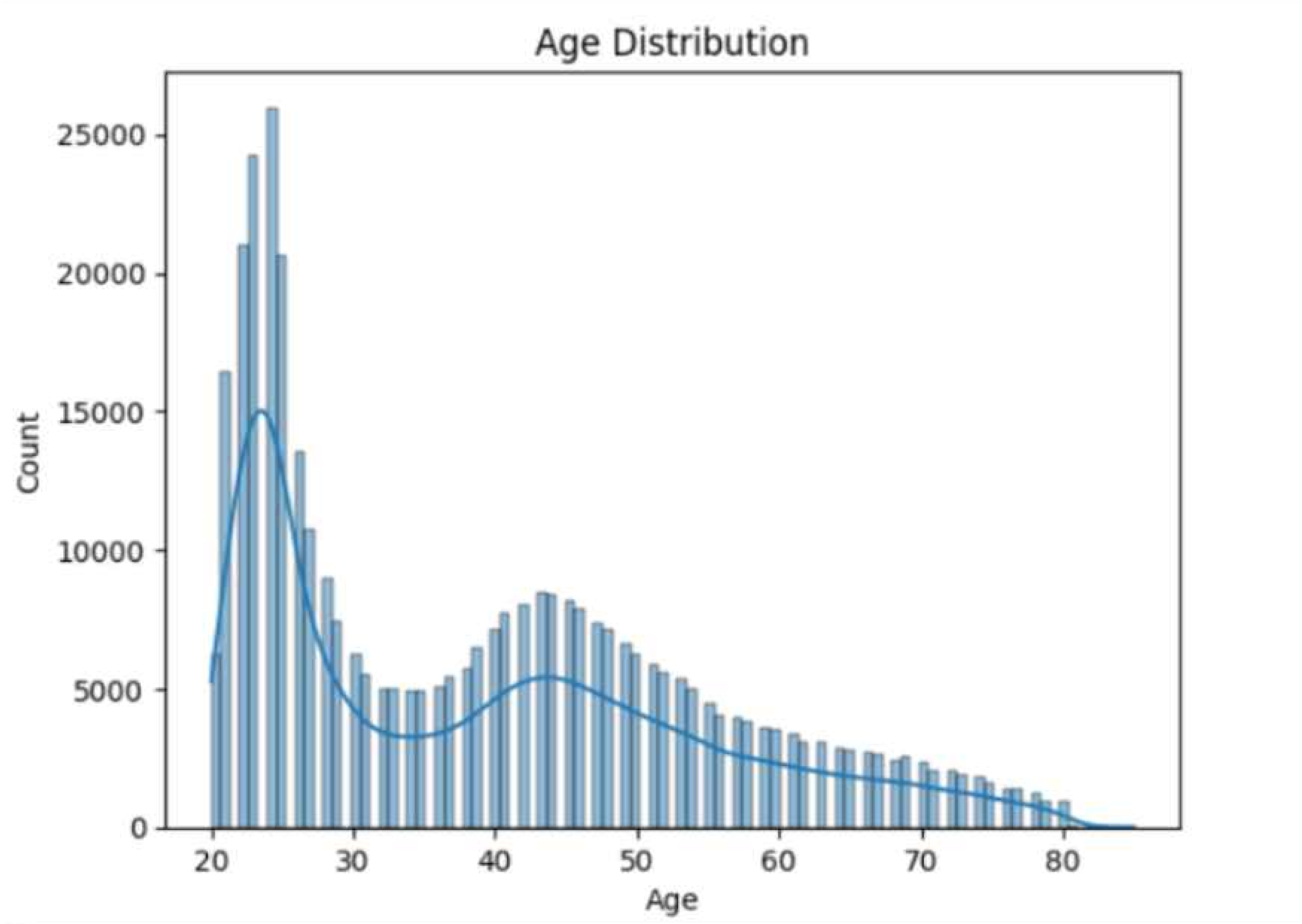
```
df.isnull().sum()
```

	0
id	0
Gender	0
Age	0
Driving_License	0
Region_Code	0
Previously_Insured	0
Vehicle_Age	0
Vehicle_Damage	0
Annual_Premium	0
Policy_Sales_Channel	0
Vintage	0
Response	0

dtype: int64

AGE DISTRIBUTION ANALYSIS

```
sns.histplot(df['Age'],kde=True)
plt.title("Age Distribution")
plt.show()
```

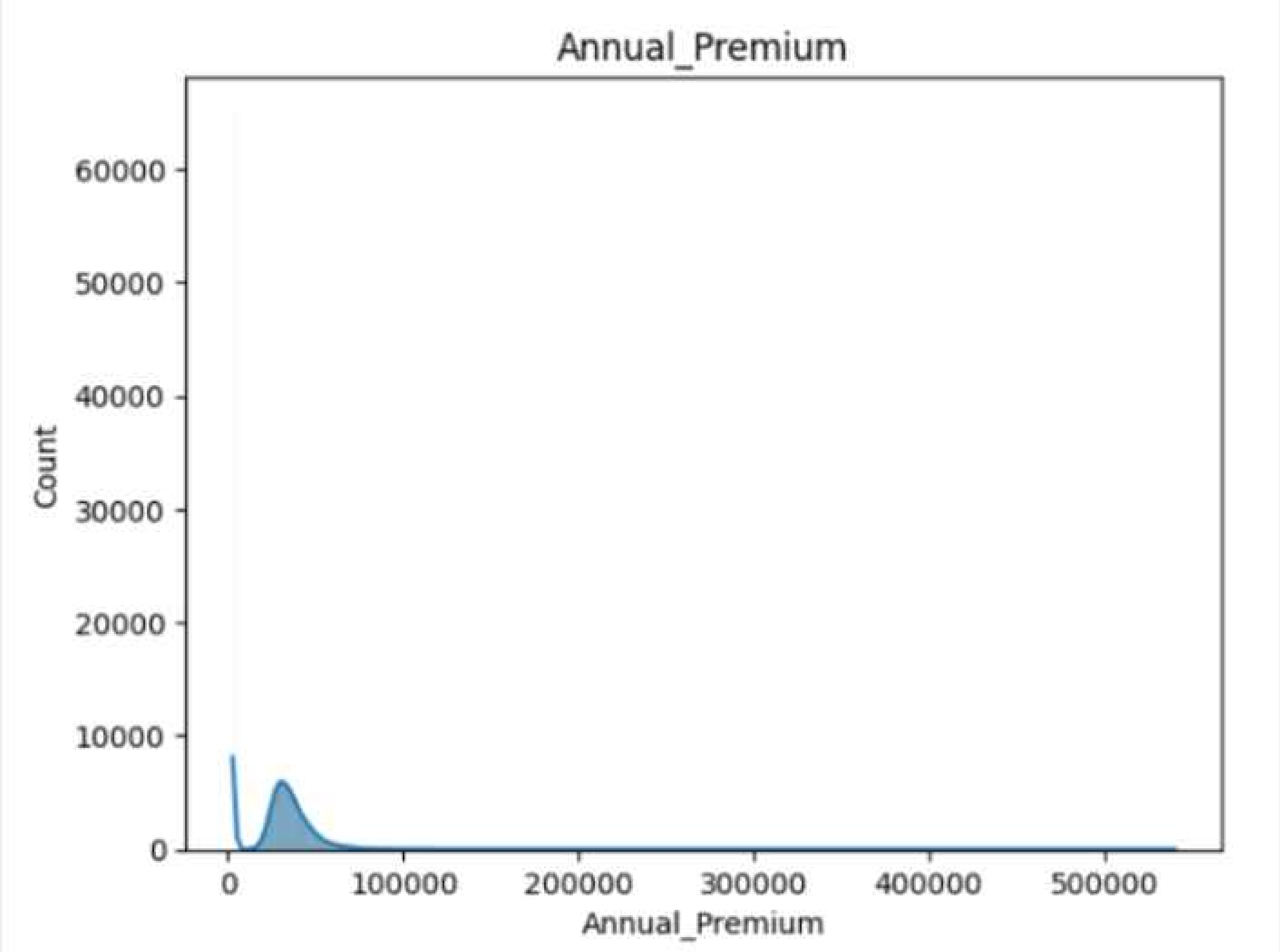


Most of the people insurance purchase between 20-30 age

CLEANER TECHNOLOGY FOR A HEALTHIER PLANET

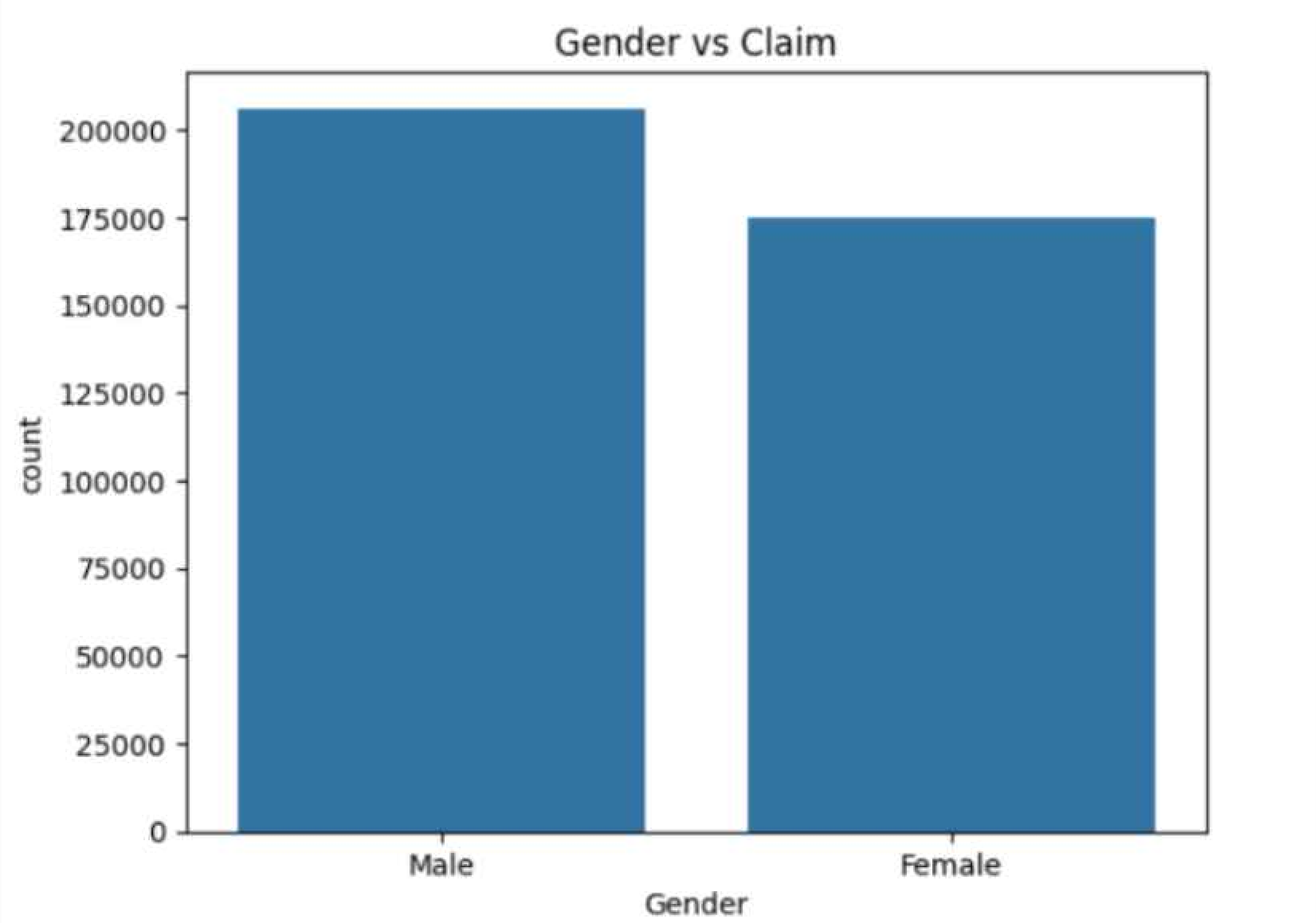
Most people have chosen the ₹10,000 annual premium plan gor their vehicle

```
sns.histplot(df["Annual_Premium"],kde=True)
plt.title("Annual_Premium")
plt.show()
```



GENDER VS CLAIM ANALYSIS

```
sns.countplot(x="Gender",data=df)
plt.title("Gender vs Claim")
plt.show()
```

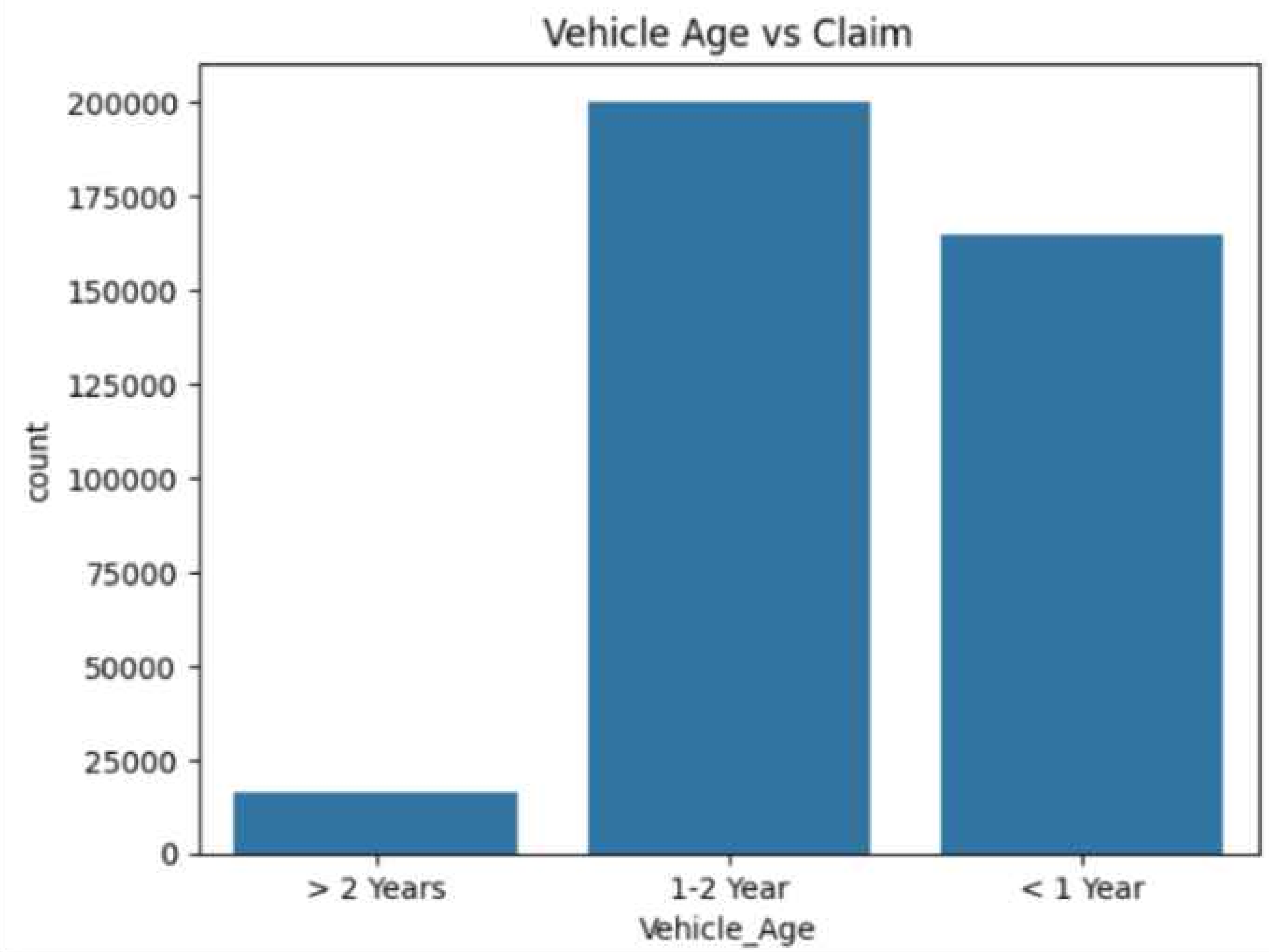


Most insurance claims were made by males then females.

VEHICLE AGE AND CLAIMS ANALYSIS

The most people have claimed their insurance within one to two years, then within one year, and after that, within two years.

```
sns.countplot(x='Vehicle_Age', data=df)
plt.title("Vehicle Age vs Claim")
plt.show()
```



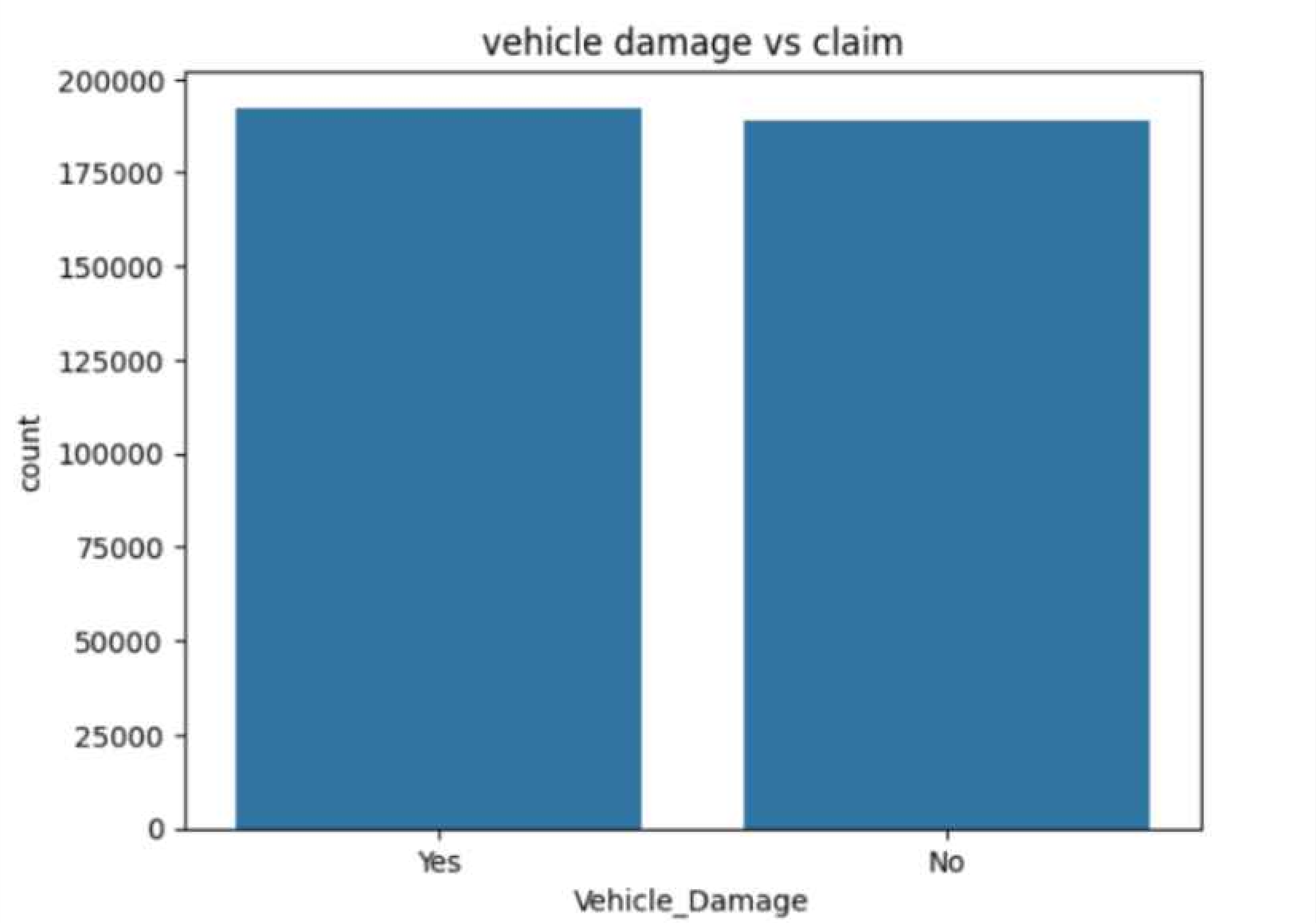
REGION-WISE ANALYSIS

```
region_claim = df.groupby('Region_Code')['Response'].mean().sort_values(ascending=False)
region_claim.head(10).plot(kind='bar')
plt.title("Top Regions by Claim Rate")
plt.show()
```

POPULAR HYBRID MODELS

CLAIM FREQUENCY BY VEHICLE DAMAGE:

```
sns.countplot(x='Vehicle_Damage',data=df)
plt.title('vehicle damage vs claim')
plt.show()
```



In vehicle damage versus claim, the number of “yes” is higher, the number of “no” is also higher, but the “yes” is just a bit more than the “no.”

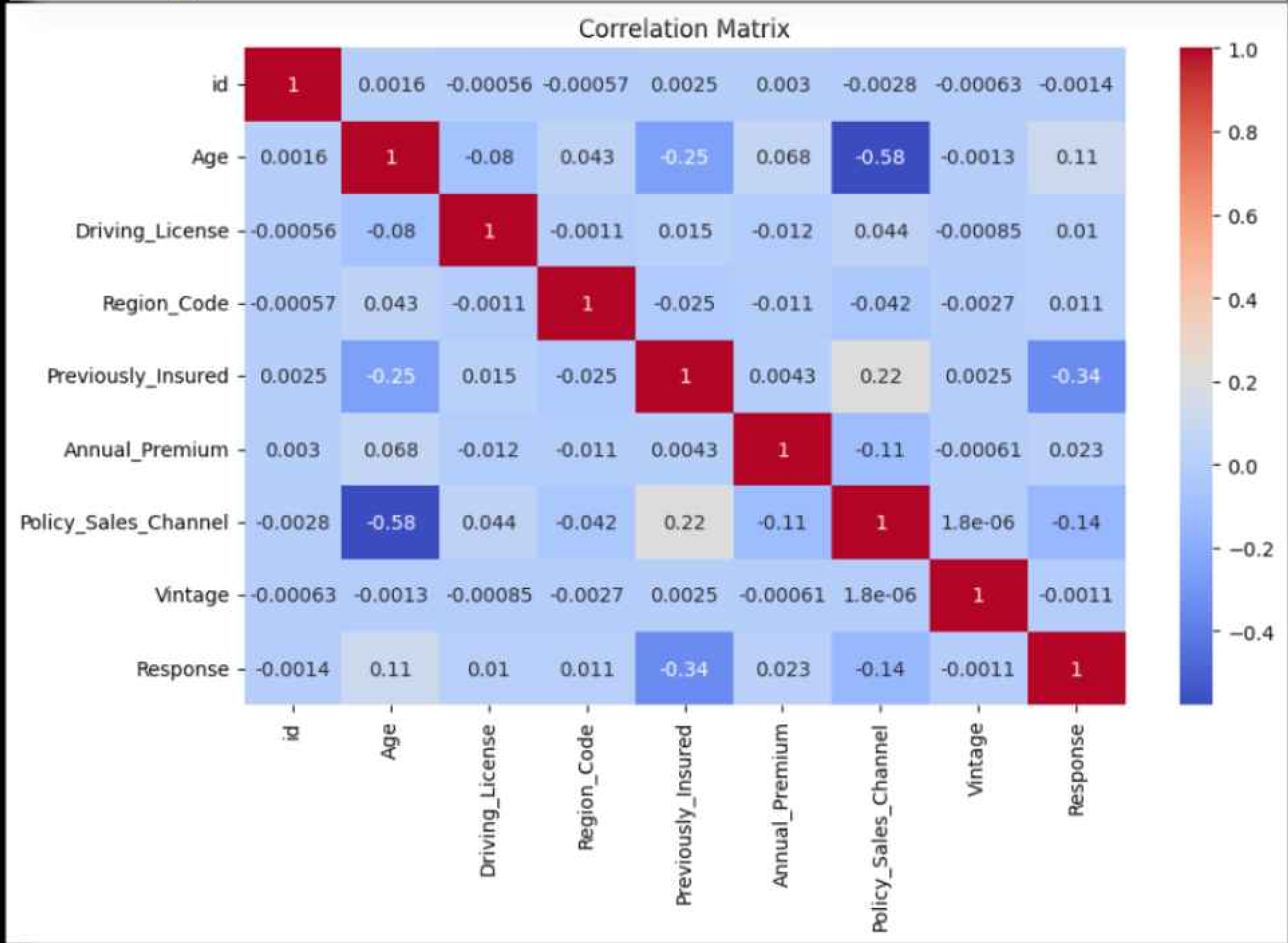
KEY INSIGHTS (PRINT SUMMARY)

```python
print("Average Claim Rate:", df['Response'].mean())
print("Average Premium:", df['Annual_Premium'].mean())
print("Average Age:", df['Age'].mean())
```

```
Average Claim Rate: 0.12256336113815208
Average Premium: 30564.389581458323
Average Age: 38.822583565331705
```

CORRELATION HEATMAP

```python
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```



CONCLUSION OF ANALYSIS

- Here's a clear and concise conclusion for your Vehicle Insurance EDA project:
- Conclusion:
- The Exploratory Data Analysis (EDA) of the vehicle insurance dataset reveals key patterns influencing policy purchases and claim behavior. Most customers prefer moderate premium plans (e.g., around 10,000 annual premium), indicating a balance between affordability and coverage. Claim frequency is higher among male customers compared to females, suggesting possible differences in driving patterns or risk exposure.
- Additionally, a significant number of claims occur within the first 1–2 years of policy purchase, highlighting a critical period for insurers to assess risk. Certain vehicle types and customer segments show higher claim tendencies, which can help companies refine pricing strategies and risk assessment models.
- Overall, the insights from EDA can help insurance companies improve customer targeting, optimize premium pricing, and reduce claim-related losses through better prediction and policy design.
-